

Challenges related to Ducts, Chimneys and Tail-end Alignment

Aligning The Tail End Of Wire Rod Coils

In Wire Rod Mill of steel plant, the end product is in coil form. The coil is formed at 550 deg C to 900 deg C by a rotating head called laying head positioned at the end of the mill line. The coil is then transferred by Stelmor conveyor for packaging. The tail end of the coil might get misaligned from the rest of the coil (it takes much bigger diameter) heap and gets entangled with the other parts of the handling device. This may get stuck in the passage. When this happens, it may end up stopping the production line, causing damage to the finished product. To avoid the above situation, currently, the tail end is manually set right by a person using a long rod while the red-hot coils are moving toward the Stelmor conveyor. The red-hot coil is placed on an intermediate conveyor moving at a speed of approximately 10 cm/s and goes into a cooling zone. After coming out of the cooling zone the coil is fed into a vertical mandrel. The front end of the coil is well placed inside the coil diameter by the laying head but the tail end comes out in a haphazard manner. Most of the time, the tail end is outside the coil loop as the conveyor drags the coil. If this is left as it is, it will get entangled while entering into the vertical mandrel. Therefore, the tail end is manually positioned every time. An operator lifts the tail end and places the tail end inside the formed coil loop. After proper placement of the tail end the operator comes out of the work zone with the tool. All these operations are done while the coil is in motion. There are two problems associated with this: It is a manual and inefficient process and it is hazardous and not very safe for the operator.